EECS 489: Final Exam, Winter 2017

Duration: 104 Minutes

OFFLINE, open-book, open-notes exam. You may consult the course textbooks, Kurose and Ross, Computer Networking: A Top-Down Approach, any edition, your own notes, notes from discussion sections, programming assignments, your solutions to them, any solutions provided by the instructors, and other notes provided by the instructors, including the course lecture slides. The above listed material may be accessed in hard copy or electronic form, offline. A calculator or a calculator program is also permitted for the exam. These are the only uses of a computer permitted. Any other use of the computer and accessing the Internet/local network is strictly forbidden. You are also not allowed to compile and run any programming code during the exam. You must not consult any resources other than those listed above.

Write legibly. If the grader cannot read what you've written, they will assume that you meant the illegible portion as a note to yourself and will ignore it. If you lose points because part of your answer is illegible, you will not be given the opportunity to explain what it says. Illegible scribble will earn zero points.

Do not ask questions during the exam. They disturb other students. Figuring out what the question is asking is part of the exam. If you think you have to make some assumption to answer a problem, note your assumption on the test. The answers to most questions should be short; you need not use all the space provided to answer the questions. If you find yourself writing an excessively long response, you may want to think more carefully about the question.

Write your uniqname on the lower-left corner of every page.

Honor code pledge. You are to abide by the University of Michigan/Engineering honor code. Sign below to indicate that you have kept the honor code pledge.

"I have neither given nor received unauthorized aid on this examination, nor have I concealed any violations of the Honor Code."

Name:

Signature:

Uniqname:

	Q1	Q2	Q3	Q4	Q5	Q6	Total
Grade	/13	/13	/12	/14	/14	/12	/78

1 [13 points] Addressing and Forwarding

(i) **[2 points]** What is the minimum number of addresses assigned to an organization in the classful addressing scheme? What would be the prefix length?

+1 256
+1 24 bit

(ii) **[3 points]** Assume that an AS has three customers with the following address prefixes: 37.3.95/24, 37.3.96/24, and 37.3.97/24. What address prefix would the AS advertise using BGP?

+3 37.3.64/18

OR

+3 for 37.3.95/24 and 37.3.96/23

(iii) **[2 points]** Michigan is building a datacenter using the fat-tree topology with $k = 8$. Meaning, it can have at most 128 physical hosts. What is the minimum possible size of the forwarding tables of its switches? What routing protocol must be used in that case?

+1 0
+1 Source routing

(iv) **[6 points]** Consider a router with 4 ports that have the following destination address range mapping:

- 200.23.16/21 $\rightarrow$ Port 0
- 200.23.24/24 $\rightarrow$ Port 1
- 200.23.24/21 $\rightarrow$ Port 2
- Otherwise $\rightarrow$ Port 3

For each of the following packet destination IP addresses, write down the port that packet will go out of using longest-prefix matching.

(A) 200.23.24.170

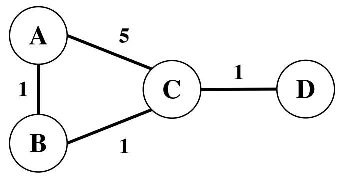
+3 Port 1

(B) 200.23.32.170

+3 Port 3

2 [13 points] Intra-Domain Routing

The distance vector (DV) routing algorithm is known to cause routing loops.



(i) **[6 points]** The network shown above uses DV, and forwarding tables of all its nodes have converged. Which *single link failure* will cause a routing loop to form (and thus lead to the count-to-infinity situation). Write down the subsequent events that led to the loop. Assume that poison reverse is not in use.

+3 A-B, B-C, or C-D

+3 A-B: loop between B and C
B-C: loop between A and B OR loop between C and D
C-D: loop between B and C

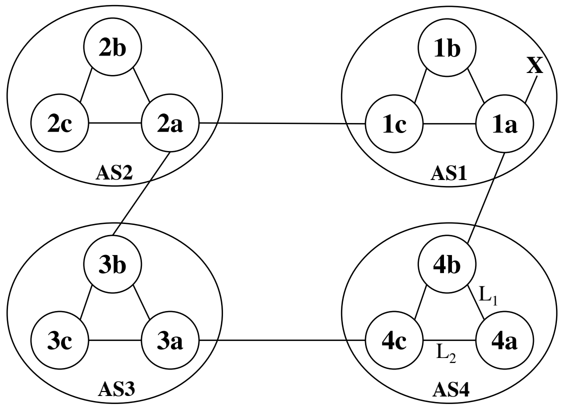
(ii) **[7 points]** Now assume that we have poison reverse enabled in this network. The idea is, if a node X uses node Y as its next hop to get to node Z, then X would advertise a cost of ∞ for node Z when sending a distance vector update message to node Y. Do you think poison reverse is sufficient to prevent the routing loop if the link between C and D fails? Explain your answer in detail.

+4 NO

+3 loop between B and C will never resolve

3 [12 points] Inter-Domain Routing

Consider the following network. Assume that all the ASes are running OSPF as their intra-AS routing protocol. Suppose eBGP and iBGP are used for the inter-AS routing protocol.



(i) **[2 points]** Router $3c$ learns about prefix X from which routing protocol?

+2 iBGP

(ii) **[2 points]** Router $1c$ learns about prefix X from which routing protocol?

+2 OSPF

(iii) **[4 points]** Once router $4a$ learns about X, it will put an entry (X, L) in its forwarding table. Will L be equal to L_1 or L_2 for this entry? Why?

+2 L1
+2 Lower ASPATH

(iv) **[4 points]** Now suppose that the link between $4b$ and $1a$ does not exist. Will the updated entry (X, L) have L equal to L_1 or L_2? Why?

+2 L2
+2 ASPATH is the same, so it goes through L2 following the shortest path to the egress router

4 [14 points] CSMA

Consider 3 hosts – A, B, and C – that are connected to a Layer-2 broadcast medium. The propagation delay between A and B is 1 ms, between B and C is also 1 ms, and between A and C is 2 ms. A has to send data to B that takes 3 ms to transmit (i.e., transmission time is 3 ms). C also has to send data to B with a transmission time of 10 ms. A starts transmitting at 0 ms and C starts at 4 ms.

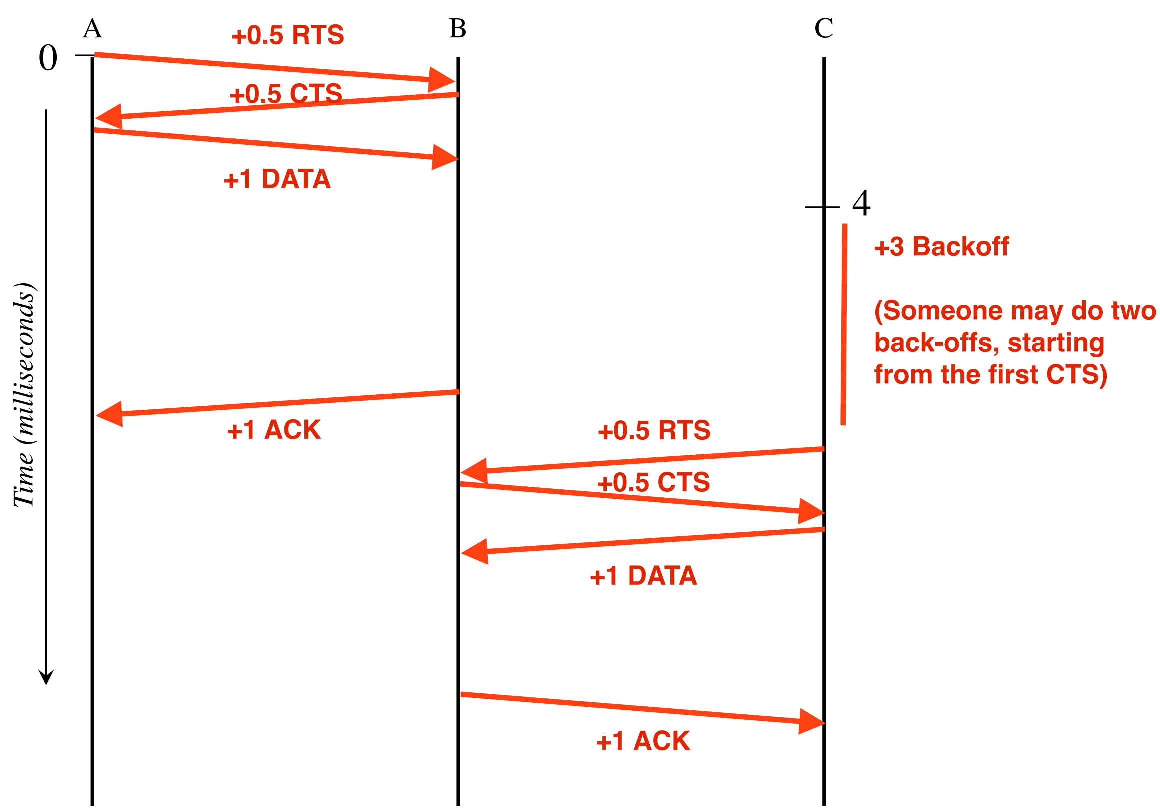
Given the broadcast medium, collisions are inevitable. A and C both use deterministic exponential back-off whenever a backoff is necessary. Specifically, A waits 2^i ms and C waits 4^i ms, where i is the i-th time a host attempts to retransmit; $i \geq 1$. For example, the first time C has to back-off for whatever reason, it backs off for 4 ms; the next back-off duration for C is 16 ms, and so on.

Assume that if a host senses carrier, it continues to listen for carrier and can transmit immediately once carrier ceases. Also assume that transmission times for all signals other than data are 0 ms.

(i) **[1 points]** Collision detection is not feasible in this medium. What is a suitable MAC protocol in such an environment?

+1 CSMA/CA

(ii) **[9 points]** Draw below all control and data message exchanges that will occur using the MAC protocol you have selected along with any required back-offs. Each exchange is a labeled arrow, where the label has the message type and the arrow starts from the source and ends at the destination for that message. Mark timestamps, but the markings need not be drawn to scale. No need to write message formats.

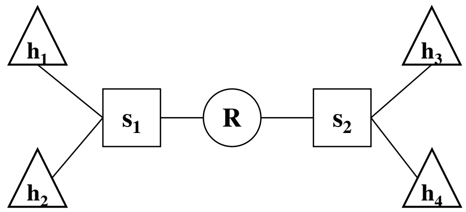


(iii) **[4 points]** What is the completion time of A's transmission? What about C's?

+2 A: 7
+2 C: Two cases
- **Single back-off: 18 or 22**
- **Double back-off: 32 or 36**

5 [14 points] Discovery

Consider the network below. The triangles represent end hosts, the squares represent layer-2 switches, and the circle represents a router.



(i) **[3 points]** Does host h_1 always broadcast ARP requests before sending a packet to h_2? Why?

+1 No
+2 ARP caching

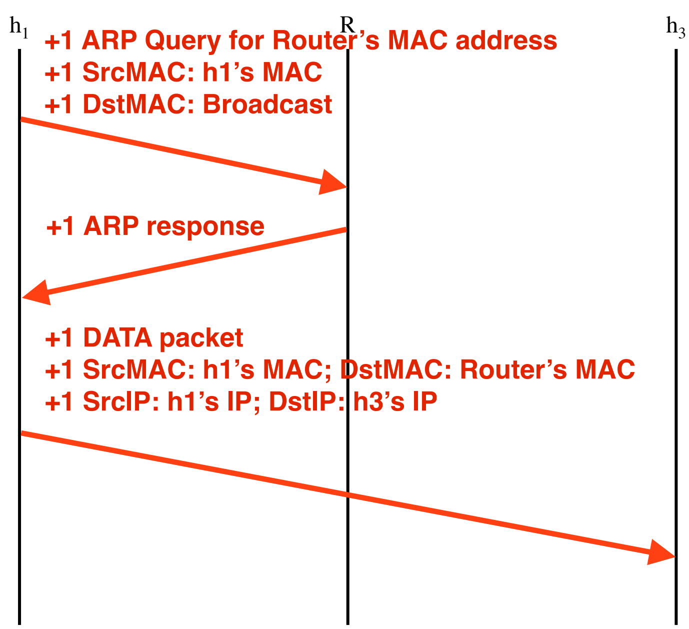
(ii) **[2 points]** What does s_1 do when it receives an ARP query?

+2 Forwards to all interfaces

(iii) **[2 points]** What does R do when it receives an ARP query?

+1 If the request is for itself then responds
+1 Otherwise drops

(iv) **[7 points]** Suppose h_1 wants to send a packet to h_3. What are the steps h_1 will perform to send the packet to h_3? Assume that all ARP caches are empty. In your answer, also include the source and destination MAC addresses for each packet. Note that it is fine to write something like "h_2's MAC address."



6 [12 points] Centralized vs. Decentralized Networking

(i) **[4 points]** What are the two main challenges of using centralization in a network?

+2 each, for any two of the following

1. **Fault-tolerance/robustness**
2. **Scalability**
3. **Security**

(ii) **[4 points]** How would you implement the Link State routing protocol (LS) in a centralized SDN-based network?

+2 Send everything to the central controller
+2 Distribute the centrally computer result everywhere

(iii) **[4 points]** Can SDN principles be used in an inter-domain scenario? If yes, give one example.

+2 Yes
+2 SDX